

Everyday Data Analysis Assignment 1: Excel

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Statement of Authentication:

By submitting this assignment, I consent that this work is my own, and where I have made use of the ideas/work of another individual, I have made appropriate acknowledgment to the original author of said work in the bibliography of this assignment.

All workings and tables can be found in the Excel file attached with this submission.

As the author, I would also like to note that there was no mention of a word count for the assignment. When I inquired with the lecturer, Johns, I was advised 800 - 1000 words. I originally prepared the assignment with that in mind.

Signed:

Chris Doyle

Signed: Christopher F. Doyle

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Introduction

The outbreak of Covid-19 caused a drastic shift in conventional workplace norms, with a departure from the traditional office environment to a system where employees are capable of working from the confines of their own home, dubbed “remote work” (Faulconer & Grusspp., 2018, pp. 156). Despite Covid-19 regulations subsiding, many employees continue to work remotely due to the perceived benefits of avoiding traffic to and from the office (Ozimek, 2020).

However, much of the literature on this subject reviews worker productivity, and if remote work has any impact on the organizational performance (Yang, *et. al.*, 2022, pp. 43). The purpose of this report is to assess the status of remote work in the global job market. In particular, if the remote work that AIB currently offers its finance data employees (a hybrid model of up to 3 days at home) is competitive with other potential firms, and to determine if there is a relationship between salaries and remote work to test if companies are using remote work as an alternative form of compensation.

Methodology

The report was conducted using a dataset by Vazquez (2023). Supplementary data was obtained from the Central Bank of Ireland (CBI, 2023) and the Financial Times (FT, 2023). A complete description of the dataset has been attached as Appendix 1.

Data Cleaning & Transformations

The feature values were checked for data quality issues (see 2.*Check Values* sheet). There were no missing values or miss-spellings but there were issues of inconsistent values in the `job_title` column. For example, some job titles based in the UK used the spelling “Data Modeller”, while their American counterparts used the spelling “Data Modeler”. These values were homogenized to the British spelling (see *cat_dq_issue* sheet). The feature `job_title_category` was created to deal with reducing the cardinality of `job_title` from 113 to 15 (see fig. 3.1).

Additionally, the `experience_level` and `employment_type` features used the short-hand versions of their values, and were replaced with the more legible values described in the documentation notes (see 4.*categorical_conversion_values* sheet). Furthermore, the remote work column used numerical values to encode values, despite only having three potential values.

For this reason, the feature was converted to categorical values using the documentation notes provided by the author (see 4.*categorical_conversion_values* sheet) and renamed “remote_work” (see figure 2). Additionally, the columns `employment_type` and `remote_work` were altered to be numbered from 1 to n , with n being the number of unique values in the feature, to reflect the fact that these are ordinal features, and make sorting them in pivot tables easier.

The only significant issue with the numerical data was that salary was in the nominal amounts of each currency rather than a single, uniform currency. There was a `salaries_in_usd` feature but the documentation did not distinguish if it used present exchange rates or the exchange rates at the time of data collection. Instead these features were converted into euro values using data provided by the CBI (2023) and Financial Times (2023) (see *6.euro_conversion_rates* sheet) using the exchange rates at the time revenue was earned as it was deemed the best practice for accurate comparison (PWC, 2023).

Other transformations include creating the `location_match` feature, which returns TRUE if the employee is resident in the same country as the head office, as it highlights where employees were able to work from abroad. Another feature, `remote_work_integrity_check` was created to check if employees were not resident to the same country as the country their employer was based in but did not have any form of remote work (I.E. where `location_match` = False and `remote_work` = “no remote work” then return False, else return True).

The intention with this feature was to detect potential logical integrity issues, but this was not necessarily the case, as employees are determined as resident from a legal perspective rather than a physical one, and the logical integrity feature is not necessarily a data quality issue where results returned are false (see further explanation in appendix 1). The final version of the data is stored in the *7.final_data sheet*.

Results:

A set of general summary statistics on each feature can be seen in the *8.statistics* sheet (see figs. 3.1-3.3). Initial results show that nearly 60% of companies do not offer the ability to work from home for more than 1 day a week (see fig 3.2), a trend that has been increasing since 2022 see (fig. 4.1), indicating that companies are attempting to move employees back into the office and as of 2023 only 55% of companies in the large category (AIB’s category) offer any form of remote work.

In terms of salary, from viewing fig. 3.3, the salaries figure was more evenly distributed, with a mean, mode and median €132k - €137k, which is an indication that the data follows a more normal distribution (Economic Research Institution (ESRI), 2023). Although Data Engineering was the most popular job title in terms of users (see fig. 3.2), fig. 5.3 shows that despite Analytics, AI and Cloud computing were on average some of the highest paid job categories.

On average, partially remote workers earned less compared to non-remote staff (€74,416.45 vs €144,749.51 respectively). At initial inspection, this supports the original hypothesis that remote work was an alternative form of compensation, but this is refuted by the fact that salaries of fully remote workers were closer to non-remote counterparts (€133,874.30 vs €144,749.51 respectively, see figs. 5.1 & 5.2).

On further analysis, this is due to experience level, as approximately 73% of all wholly remote and non-remote staff are senior level managers, while 71% of all partial remote staff are either entry level or intermediate level staff (see figs. 6.1, 6.2 & 6.3).

Limitations

This report does not look at the impact of remote work from a social or mental health perspective, as this data can be hard to quantify in numerical terms, and involves obtaining data regarded as highly sensitive under GDPR (MHC, 2023). Furthermore, the data does not reflect the cost of living between different countries and where remote workers may be earning more or less depending on their location of residence.

Attempts were made to incorporate information such as exchange rates at the time of employment, but cost of living is a feature that could be implemented into the report if more time and resources were available. Furthermore, there were only 7 data points (see *11.irish_data* sheet), meaning it is difficult to assess AIB with its domestic competitors, and this data specifically reflects jobs within the technology industry. Although this is applicable for the Finance Data teams and similar departments, it may not be as beneficial for those working in other departments such as Accounting or HR.

Conclusion and Recommendations

The 3 day remote work system that AIB offers is significantly more competitive than its global counterparts. Although more investigation is needed into the relationship between remote work and remuneration, evidence suggests that experience level has a more significant influence on remuneration than if employees are remote or not. Additional research is required to build cost of living into these analyses,

Appendix 1: Dataset Description

A high level description of the features listed in the dataset provided by Vazquez (2023), its data type (in brackets), and a more detailed description of the data quality issues identified and checked for during the course of this assignment.

Work_year (numeric, discrete): The year in which the employee was employed by the company. There were no missing values in this feature, nor any logical integrities with the data (e.g. the presence of years 2024 or later). No data cleaning was required for this feature.

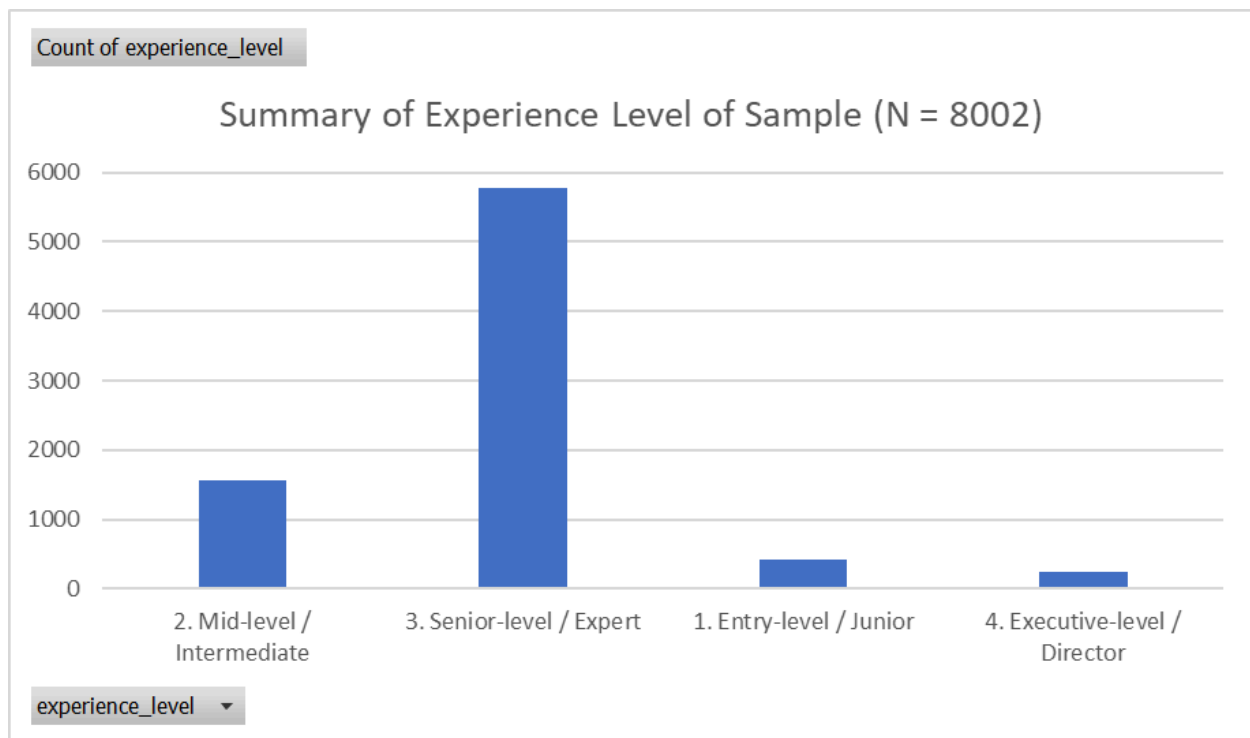
A breakdown of the values can be found below (see *9.1.categorical_features* sheet for all graphs listed in the appendix unless otherwise stated).

Year	
Row Labels	Count of work_year
2020	75
2021	218
2022	1650
2023	6059
Grand Total	8002

The majority of data points occur within 2023. For this reason, any metric that assesses changing values over time will use % amounts related to that year, rather than the nominal amounts themselves, as it might show increasing trends when there otherwise may not be the case.

Experience_level (categorical, ordinal): A feature that reflects the proficiency/years of experience of the employee within their field of work. As stated in the report, the original values within this feature were abbreviated, but were replaced with their unabbreviated values (see table below). Additionally, each value was given a number (1 - 4) to reflect the original scaling of the feature (i.e. Entry level is lower level to Mid-level, Mid-level is lower to Senior, etc).

Original Values	New Values
EN	1. Entry-level / Junior
MI	2. Mid-level / Intermediate
SE	3. Senior-level / Expert
EX	4. Executive-level / Director



The majority of values occur within the Mid-level/Intermediate and the Senior-level/Expert categories (over 90%), although the Executive/Director roles within companies are much rarer, thus it is to be expected to be the smallest data type.

Employment_type (categorical, nominal) : The type of contract held by the employee. Similar to experience level, these were originally written in abbreviations of their true values, and were amended to their intended values, as per their original descriptions in the documentation.

Original Values	New Values
PT	Part-time
FT	Full-time
CT	Contract
FL	Freelance

Employment Type

Row Labels	Count of employment_type
Contract	18
Freelance	11
Full-time	7960
Part-time	13
Grand Total	8002

Over 98% of data points are Full-Time contracts. This will make it difficult to perform analysis when comparing other features by Employment type, as there is very little data outside of Full-Time employment.

Job_title (categorical, nominal): The role the employee held during the year. This feature was extremely long, and so the table could not be included in the report (*9.1.categorical_features* sheet). As per the data quality issues identified in the report, some features had multiple values despite having the same intended meaning. A table has been provided below to document the exact changes (see *3.cat_dq_issue* sheet).

Issue 1: Data Modeler VS Data Modeller				
Count of Job Title per Location		Column Labels		
Row Labels	GB	US	Grand Total	
Data Modeler	2	18	20	
Data Modeller	2		2	
Grand Total	4	18	22	
Issue 2: Bi VS Business Intelligence				
Row Labels	Count of Job Title			
BI Analyst	23			
BI Data Analyst	16			
BI Data Engineer	1			
BI Developer	42			
Business Intelligence Analyst	32			
Business Intelligence Data Analys	2			
Business Intelligence Developer	15			
Business Intelligence Engineer	104			
Grand Total	235			

Table one documents the misspelling between data “modeller” and “modeler”, while the second table documents how there were many instances where “Business Intelligence” was abbreviated to “BI”. This was corrected for the final submission (WARNING: These changes were made manually by hand. Do not attempt to hit “REFRESH ALL” in the workbook, as that will cause the workbook to break.

Job_title_category (categorical, nominal): A field derived from job_title. Shows the general area a job is based in. For example, the value of “machine learning engineer” under job_title would translate to “machine learning” under job_title_category. There was a great deal of subjectivity used in this section, and any job title where it was vague as to what category a role should fall under, it was put down as Data (Unspecified). A full breakdown of the conversion of values from job_title to job_title_category can be found on the *4.categorical_conversion_values* sheet.

Job Title Category	
Row Labels	Count of job_title_category
Data Science	1929
Data Engineer	1929
Analytics	1541
Machine Learning	1126
Research	368
Data Management	349
Science (Unspecified)	262
Business Intelligence	236
Data (Unspecified)	89
AI	78
Data Operations	32
Data Quality	15
Natural Language Processing	13
Cloud Computing	13
Big Data	12
Data Visualization	10
Grand Total	8002

The majority of jobs fall into the Data Science and Data Engineer category. Interesting, Data Engineer was also the most popular

Salary: (numeric, continuous): The total gross salary amount paid in the nominal amount of the country's currency during the course of that year. This feature is completely unusable without a uniform currency to compare. This feature and the salary_currency feature as well as a significant amount of data provided by the CBI (2023) and Financial Times (2023) to generate the salaries_in_eur feature.

Salary_currency (categorical, nominal): The currency the employee was paid in. It uses the shorthand name of the currency. Transformations have been applied to account for the most recent currency translation values using the data provided by the Central Bank of Ireland (2023) and Financial Times (2023)

Salary_in_usd (numeric, continuous): The salary converted to United States Dollar (USD). It is not stated as whether or not the data used is using present data conversion rates, or conversion rates at the time the data was collected. For this reason, the feature was dropped in favor of salary_in_eur.

Employee_residence (categorical, nominal): Employee's primary country of residence during the work year. There were no significant issues with the data within this feature, and no data correction techniques were required.

Remote_ratio/remote_work (categorical, ordinal): The overall amount of work done remotely, possible values are as follows: Originally encountered as a numerical feature, it is better reflected as a categorical feature with an ordinal scaling, and similar to experience-level, the following corrections were made to its values (see table below):

Remote_ratio (Old)	Remote_work (New)
0	1. No remote work (less than 20%)
50	2. Partially remote/hybrid
100	3. Fully remote (more than 80%)

Company_location (categorical, nominal): The country of the employer's main office or contracting branch is located. The top 7 most popular employee_residence and company_location are contained below:

Employee Residence		Company Location	
Row Labels	Count of employee_residence	Row Labels	Count of company_location
US	6796	US	6843
GB	392	GB	399
CA	180	CA	181
ES	109	ES	106
DE	65	DE	72
IN	64	IN	50
FR	51	FR	47

The US, UK and Canada are the most popular in terms of employee and Company residence. This will make it difficult to compare AIB with its Irish competitors.

Company_size (categorical, ordinal): The metric for tracking the size of the company based on the average number of people that worked for the company during the year. Although this feature was abbreviated, it was decided the abbreviated values (S, M and L) were acceptable values and were not converted to their unabbreviated values, with are included below:

- S: less than 50 employees (small)
- M: 50 to 250 employees (medium)
- L: more than 250 employees (large)

The vast majority of data falls into the medium company size (approximately 89%). This will make it difficult to compare figures based on company size, as there is limited data on both small and large companies.

Company Size	
Row Labels	Count of company_size
L	702
M	7134
S	166
Grand Total	8002

Location_match (numeric, boolean): Returns True if the employee is a resident of the country that the company's main office/contracting branch is located (ie employee_residence = company_location) else returns false.

Salary_in_euro (numeric, discrete): Returns the values in Salary converted to Euros using the conversion rates included in the excel workbook. A series of statistical measures can be seen below:

Numerical Values	Cardinality	Mode	Mean	Median	Max	Min	Standard Deviation
salary	1341	150,000.00	176,635.15	145,000.00	30,400,000.00	14,000.00	474,901.51
salary_in_eur	1881	138,752.73	138,650.29	132,947.33	410,431.60	12,682.41	59,406.58

Numerical Values	Mode Count	Mode Count (%)	Median Count	Median Count (%)
salary	179	2%	109	1%
salary_in_eur	121	2%	23	0%

The salaries were evenly distributed, with a mean, mode and median in the range of €132k - €137k. Normally with salaries, the mean is prone to being skewed by extremely high values (ESRI, 2023). This indicates that salaries in this field of work are surprisingly more equal than anticipated.

Remote_work_integrity_check (numeric, boolean): Indicates where an employee is not a resident in the country they are working and are not allowed to work remotely. Originally this was a logic check to make sure that there weren't any instance where location_match and remote_work were both false (indicating that the employee wasn't a resident of the country in question but remote work wasn't allowed) but this is not necessarily a data quality issue but a potential case where the employee is an expat/migrant worker and has yet to qualify for residency at the time of data collection. For this reason, it was advised not to drop any data as a result of this issue, but the feature was included in this appendix to document that this logical integrity issue was considered in the data cleaning process.

Appendix 2: Figures

Figure 1: Example of Inconsistent-Spelling (see *cat_dq_issue* sheet in the excel file)

Count of company_location		Column Labels	
Row Labels		GB	US
Data Modeler		2	18
Data Modeller		2	2
Grand Total		4	18

Figure 2: New Values for the remote_work Column (see *4.categorical_conversion_values* sheet in the excel file)

Remote Work	
Row Labels	Count of remote_work
1. No remote work	4695
2. Partially remote/hybrid	216
3. Fully remote (more than 80%)	3091
Grand Total	8002

Figure 3.1 : General Summary Statistics (see *8.statistics* sheet in the excel file)

Dataset	
Row Count	14
Column Count	8002

Figure 3.2 : General Summary Statistics on Categorical Features (see *8.statistics* sheet in the excel file)

Categorical Features	Cardinality	Mode	Count	Count%
work_year	4	2023	6059	76%
experience_level	4	3. Senior-level / Expert	5772	72%
employment_type	4	Full-time	7960	99%
job_title	113	Data Engineer	1900	24%
job_title_category	16	Data Engineer	1929	24%
salary_currency	22	USD	7242	91%
employee_residence	85	US	6796	85%
remote_work	3	1. No remote work	4695	59%
company_location	73	US	6843	86%
company_size	3	M	7134	89%
location_match	2	TRUE	7883	99%

Figure 3.3 : General Summary Statistics on Numerical Features (see 8.statistics sheet in the excel file)

Numerical Values	Cardinality	Mode	Mean	Median	Max	Min	Standard Deviation
salary	1341	150,000.00	176,635.15	145,000.00	30,400,000.00	14,000.00	474,901.51
salary_in_eur	1881	138,752.73	138,650.29	132,947.33	410,431.60	12,682.41	59,406.58

Numerical Values	Mode Count	Mode Count (%)	Median Count	Median Count (%)
salary	179	2%	109	1%
salary_in_eur	121	2%	23	0%

Figure 4.1: Percentage of Remote Work availability Over Time (see 9.3.remote_work_analysis sheet in the excel file)

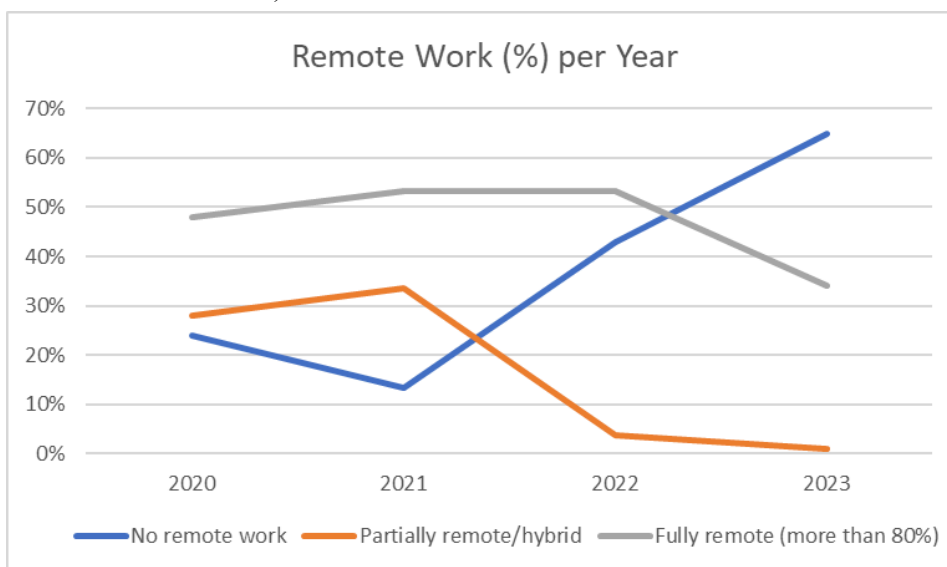


Figure 4.2: Remoter Work (%) Per Company Size

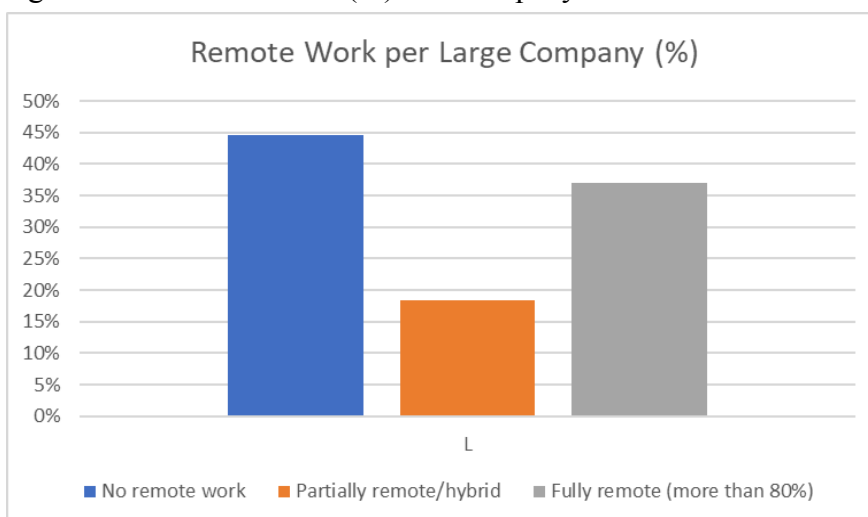


Figure 5.1: Earnings per Remote Work (see 10.1.salary_analysis sheet)

Row Labels	Average of salary_in_eur	Min of salary_in_eur	Max of salary_in_eur	StdDev of salary_in_eur
1. No remote work	144,749.51	13,132.60	410,431.60	59,476.82
1. Entry-level / Junior	98,081.86	14,244.36	231,994.56	44,899.10
2. Mid-level / Intermediate	116,472.19	15,218.89	410,431.60	53,744.51
3. Senior-level / Expert	153,615.11	15,992.75	378,298.36	57,906.39
4. Executive-level / Director	167,319.21	13,132.60	326,716.43	63,013.63
2. Partially remote/hybrid	74,416.45	14,295.65	373,360.72	55,506.82
1. Entry-level / Junior	59,599.02	16,135.48	218,876.60	42,646.58
2. Mid-level / Intermediate	69,143.07	14,295.65	357,644.02	52,039.57
3. Senior-level / Expert	94,828.10	17,122.68	373,360.72	67,182.23
4. Executive-level / Director	120,467.78	70,000.00	194,463.65	46,400.22
3. Fully remote (more than 80%)	133,874.70	12,682.41	384,597.62	56,410.69
1. Entry-level / Junior	68,155.70	12,682.41	234,020.35	40,802.62
2. Mid-level / Intermediate	95,406.87	14,244.36	314,506.19	45,917.36
3. Senior-level / Expert	146,346.22	14,587.95	384,597.62	49,846.58
4. Executive-level / Director	189,629.47	50,876.00	351,725.56	68,008.86
Grand Total	138,650.29	12,682.41	410,431.60	59,406.58

Figure 5.2: Average Salary by Remote Work (see 10.2.salary_analysis sheet)

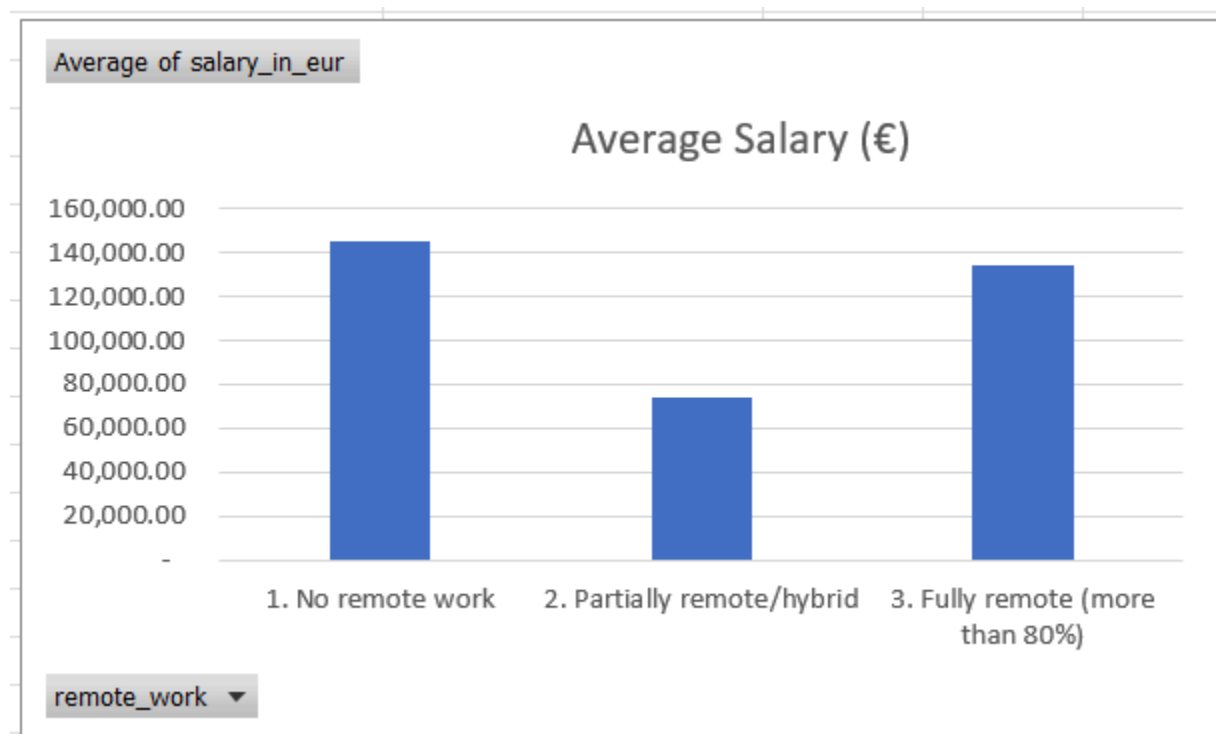


Figure 5.3: Highest Average Salaries by Job_Title_Category (see 10.1.salary_analysis sheet).

Highest Paid Among	Amount	Category
1. Entry-level / Junior	154,807.96	Science (Unspecified)
2. Mid-level / Intermediate	157,517.69	AI
3. Senior-level / Expert	179,104.21	Cloud Computing
4. Executive-level / Director	194,962.38	Analytics

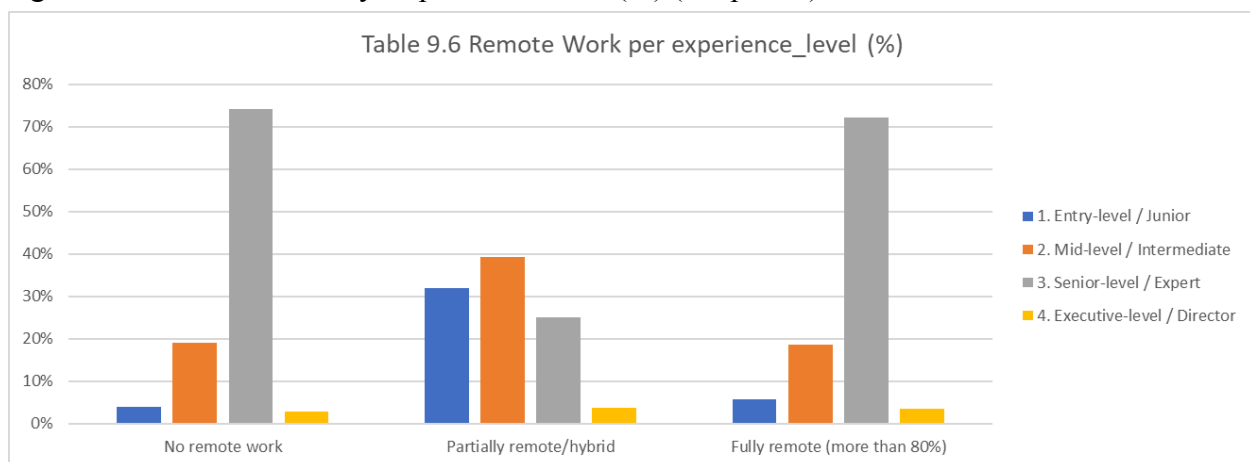
Figure 6.1: Remote Work per Contract Type (see 9.3.remote_work_analysis sheet)

Table 9.5 Remote Work per Contract Type				
Count of remote_work	Column Labels			
Row Labels	1. No remote work	2. Partially remote/hybrid	3. Fully remote (more than 80%)	Grand Total
1. Entry-level / Junior	183.00	69.00	177.00	429.00
2. Mid-level / Intermediate	895.00	85.00	576.00	1,556.00
3. Senior-level / Expert	3,487.00	54.00	2,231.00	5,772.00
4. Executive-level / Director	130.00	8.00	107.00	245.00
Grand Total	4,695.00	216.00	3,091.00	8,002.00

Figure 6.2: Remote Work By Experience Level (%)

Remote Work Type (%) With	1. Entry-level / Junior	2. Mid-level / Intermediate	3. Senior-level / Expert	4. Executive-level / Director
No remote work	4%	19%	74%	3%
Partially remote/hybrid	32%	39%	25%	4%
Fully remote (more than 80%)	6%	19%	72%	3%

Figure 6.3: Remote Work By Experience Level (%) (Graphical)



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